logistic — Logistic regression, reporting odds ratios

Description | Quick start | Menu | Syntax | Options
Remarks and examples | Stored results | Methods and formulas | References | Also see

Description

`logistic` fits a logistic regression model of *depvar* on *indepvars*, where *depvar* is a 0/1 variable (or, more precisely, a 0/non-0 variable). Without arguments, `logistic` redisplays the last `logistic` estimates. `logistic` displays estimates as odds ratios; to view coefficients, type `logit` after running `logistic`. To obtain odds ratios for any covariate pattern relative to another, see [R] **lincom**.

Quick start

Report odds ratios from logistic regression of `y` on `x1` and `x2`

```
logistic y x1 x2
```

Add indicators for values of categorical variable `a`

```
logistic y x1 x2 i.a
```

Same as above, and apply frequency weights defined by `wvar`

```
logistic y x1 x2 i.a [fweight=wvar]
```

Show base level of `a`

```
logistic y x1 x2 i.a, baselevels
```

Menu

Statistics > Binary outcomes > Logistic regression

Syntax

```
logistic depvar indepvars [if] [in] [weight] [, options]
```

options	Description
Model	
noconstant	suppress constant term
offset(*varname*)	include *varname* in model with coefficient constrained to 1
asis	retain perfect predictor variables
constraints(*constraints*)	apply specified linear constraints
SE/Robust	
vce(*vcetype*)	*vcetype* may be oim, opg, robust, cluster *clustvar*, bootstrap, or jackknife
Reporting	
level(#)	set confidence level; default is level(95)
coef	report estimated coefficients
nocnsreport	do not display constraints
display_options	control columns and column formats, row spacing, line width, display of omitted variables and base and empty cells, and factor-variable labeling
Maximization	
maximize_options	control the maximization process; seldom used
collinear	keep collinear variables
coeflegend	display legend instead of statistics

indepvars may contain factor variables; see **[U] 11.4.3 Factor variables**.

depvar and *indepvars* may contain time-series operators; see **[U] 11.4.4 Time-series varlists**.

bayes, bayesboot, bootstrap, by, collect, fp, jackknife, mfp, mi estimate, nestreg, rolling, statsby, stepwise, and svy are allowed; see **[U] 11.1.10 Prefix commands**. For more details, see [BAYES] **bayes: logistic**.

vce(bootstrap) and vce(jackknife) are not allowed with the mi estimate prefix; see [MI] **mi estimate**.

Weights are not allowed with the bootstrap prefix; see [R] **bootstrap**.

vce() and weights are not allowed with the svy prefix; see [SVY] **svy**.

fweights, iweights, and pweights are allowed; see **[U] 11.1.6 weight**.

collinear and coeflegend do not appear in the dialog box.

See **[U] 20 Estimation and postestimation commands** for more capabilities of estimation commands.

Options

Model

noconstant, offset(*varname*), constraints(*constraints*); see [R] **Estimation options**.

asis forces retention of perfect predictor variables and their associated perfectly predicted observations and may produce instabilities in maximization; see [R] **probit**.

SE/Robust

vce(*vcetype*) specifies the type of standard error reported, which includes types that are derived from asymptotic theory (oim, opg), that are robust to some kinds of misspecification (robust), that allow for intragroup correlation (cluster *clustvar*), and that use bootstrap or jackknife methods (bootstrap, jackknife); see [R] ***vce_option***.

Reporting

level(#); see [R] **Estimation options**.

coef causes logistic to report the estimated coefficients rather than the odds ratios (exponentiated coefficients). coef may be specified when the model is fit or may be used later to redisplay results. coef affects only how results are displayed and not how they are estimated.

nocnsreport; see [R] **Estimation options**.

display_options: noci, nopvalues, noomitted, vsquish, noemptycells, baselevels, allbaselevels, nofvlabel, fvwrap(#), fvwrapon(*style*), cformat(%*fmt*), pformat(%*fmt*), sformat(%*fmt*), and nolstretch; see [R] **Estimation options**.

Maximization

maximize_options: difficult, technique(*algorithm_spec*), iterate(#), [no]log, trace, gradient, showstep, hessian, showtolerance, tolerance(#), ltolerance(#), nrtolerance(#), nonrtolerance, and from(*init_specs*); see [R] **Maximize**. These options are seldom used.

The following options are available with logistic but are not shown in the dialog box:

collinear, coeflegend; see [R] **Estimation options**.

Remarks and examples

Remarks are presented under the following headings:

logistic and logit
Robust estimate of variance
Video examples

logistic and logit

logistic provides an alternative and preferred way to fit maximum-likelihood logit models, the other choice being logit ([R] **logit**).

First, let's dispose of some confusing terminology. We use the words logit and logistic to mean the same thing: maximum likelihood estimation. To some, one or the other of these words connotes transforming the dependent variable and using weighted least squares to fit the model, but that is not how we use either word here. Thus, the logit and logistic commands produce the same results.

The logistic command is generally preferred to the logit command because logistic presents the estimates in terms of odds ratios rather than coefficients. To some people, this may seem disadvantageous, but you can type logit without arguments after logistic to see the underlying coefficients. You should be cautious when interpreting the odds ratio of the constant term. Usually, this odds ratio represents the baseline odds of the model when all predictor variables are set to zero. However, you must verify that a zero value for all predictor variables in the model actually makes sense before continuing with this interpretation.

Nevertheless, [R] **logit** is still worth reading because logistic shares the same features as logit, including omitting variables due to collinearity or one-way causation.

For an introduction to logistic regression, see Lemeshow and Hosmer (2005), Pagano and Gauvreau (2022, 455–478), or Pampel (2021); for a complete but nonmathematical treatment, see Kleinbaum and Klein (2010); and for a thorough discussion, see Hosmer, Lemeshow, and Sturdivant (2013). See Gould (2000) for a discussion of the interpretation of logistic regression. See Dupont (2009) or Hilbe (2009) for a discussion of logistic regression with examples using Stata. For a discussion using Stata with an emphasis on model specification, see Vittinghoff et al. (2012).

Stata has a variety of commands for performing estimation when the dependent variable is dichotomous or polytomous. See Long and Freese (2014) for a book devoted to fitting these models with Stata. See help estimation commands for a complete list of all of Stata's estimation commands.

▷ Example 1

Consider the following dataset from a study of risk factors associated with low birthweight described in Hosmer, Lemeshow, and Sturdivant (2013, 24).

```
. use https://www.stata-press.com/data/r19/lbw
(Hosmer & Lemeshow data)
. describe
Contains data from https://www.stata-press.com/data/r19/lbw.dta
 Observations:           189                  Hosmer & Lemeshow data
    Variables:            11                  15 Jan 2024 05:01
```

Variable name	Storage type	Display format	Value label	Variable label
id	int	%8.0g		Identification code
low	byte	%8.0g		Birthweight<2500g
age	byte	%8.0g		Age of mother
lwt	int	%8.0g		Weight at last menstrual period
race	byte	%8.0g	race	Race
smoke	byte	%9.0g	smoke	Smoked during pregnancy
ptl	byte	%8.0g		Premature labor history (count)
ht	byte	%8.0g		Has history of hypertension
ui	byte	%8.0g		Presence, uterine irritability
ftv	byte	%8.0g		Number of visits to physician during 1st trimester
bwt	int	%8.0g		Birthweight (grams)

```
Sorted by:
```

We want to investigate the causes of low birthweight. Here race is a categorical variable indicating whether a person is white (race = 1), black (race = 2), or some other race (race = 3). We want indicator (dummy) variables for race included in the regression, so we will use factor variables.

```
. logistic low age lwt i.race smoke ptl ht ui
Logistic regression                                     Number of obs =    189
                                                        LR chi2(8)    =  33.22
                                                        Prob > chi2   = 0.0001
Log likelihood = -100.724                               Pseudo R2     = 0.1416

------------------------------------------------------------------------------
         low | Odds ratio   Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         age |   .9732636   .0354759    -0.74   0.457     .9061578    1.045339
         lwt |   .9849634   .0068217    -2.19   0.029     .9716834    .9984249
             |
        race |
       Black |   3.534767   1.860737     2.40   0.016     1.259736    9.918406
       Other |   2.368079   1.039949     1.96   0.050     1.001356    5.600207
             |
       smoke |   2.517698    1.00916     2.30   0.021     1.147676    5.523162
         ptl |   1.719161   .5952579     1.56   0.118     .8721455    3.388787
          ht |   6.249602   4.322408     2.65   0.008     1.611152    24.24199
          ui |     2.1351   .9808153     1.65   0.099     .8677528      5.2534
       _cons |   1.586014   1.910496     0.38   0.702     .1496092     16.8134
------------------------------------------------------------------------------
Note: _cons estimates baseline odds.
```

The odds ratios are for a one-unit change in the variable. If we wanted the odds ratio for age to be in terms of 4-year intervals, we would type

```
. generate age4 = age/4
. logistic low age4 lwt i.race smoke ptl ht ui
(output omitted)
```

After logistic, we can type logit to see the model in terms of coefficients and standard errors:

```
. logit
Logistic regression                                     Number of obs =    189
                                                        LR chi2(8)    =  33.22
                                                        Prob > chi2   = 0.0001
Log likelihood = -100.724                               Pseudo R2     = 0.1416

------------------------------------------------------------------------------
         low | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
        age4 |  -.1084012   .1458017    -0.74   0.457    -.3941673    .1773649
         lwt |  -.0151508   .0069259    -2.19   0.029    -.0287253   -.0015763
             |
        race |
       Black |   1.262647   .5264101     2.40   0.016     .2309024    2.294392
       Other |   .8620792   .4391532     1.96   0.050     .0013548    1.722804
             |
       smoke |   .9233448   .4008266     2.30   0.021      .137739    1.708951
         ptl |   .5418366    .346249     1.56   0.118     -.136799    1.220472
          ht |   1.832518   .6916292     2.65   0.008     .4769494    3.188086
          ui |   .7585135   .4593768     1.65   0.099    -.1418484    1.658875
       _cons |   .4612239    1.20459     0.38   0.702    -1.899729    2.822176
------------------------------------------------------------------------------
```

If we wanted to see the logistic output again, we would type logistic without arguments.

◁

▷ Example 2

We can specify the confidence interval for the odds ratios with the `level()` option, and we can do this either at estimation time or when replaying the model. For instance, to see our first model in example 1 with narrower, 90% confidence intervals, we might type

```
. logistic, level(90)
Logistic regression                                     Number of obs =    189
                                                        LR chi2(8)    =  33.22
                                                        Prob > chi2   = 0.0001
Log likelihood = -100.724                               Pseudo R2     = 0.1416
```

low	Odds ratio	Std. err.	z	P>|z|	[90% conf.	interval]
age4	.8972675	.1308231	-0.74	0.457	.7059409	1.140448
lwt	.9849634	.0068217	-2.19	0.029	.9738063	.9962483
race						
Black	3.534767	1.860737	2.40	0.016	1.487028	8.402379
Other	2.368079	1.039949	1.96	0.050	1.149971	4.876471
smoke	2.517698	1.00916	2.30	0.021	1.302185	4.867819
ptl	1.719161	.5952579	1.56	0.118	.9726876	3.038505
ht	6.249602	4.322408	2.65	0.008	2.003487	19.49478
ui	2.1351	.9808153	1.65	0.099	1.00291	4.545424
_cons	1.586014	1.910496	0.38	0.702	.2186791	11.50288

Note: **_cons** estimates baseline odds.

◁

Robust estimate of variance

If you specify `vce(robust)`, Stata reports the robust estimate of variance described in [U] **20.22 Obtaining robust variance estimates**. Here is the model previously fit with the robust estimate of variance:

```
. logistic low age lwt i.race smoke ptl ht ui, vce(robust)
Logistic regression                                     Number of obs =    189
                                                        Wald chi2(8)  =  29.02
                                                        Prob > chi2   = 0.0003
Log pseudolikelihood = -100.724                         Pseudo R2     = 0.1416
```

low	Odds ratio	Robust std. err.	z	P>|z|	[95% conf.	interval]
age	.9732636	.0329376	-0.80	0.423	.9108015	1.040009
lwt	.9849634	.0070209	-2.13	0.034	.9712984	.9988206
race						
Black	3.534767	1.793616	2.49	0.013	1.307504	9.556051
Other	2.368079	1.026563	1.99	0.047	1.012512	5.538501
smoke	2.517698	.9736417	2.39	0.017	1.179852	5.372537
ptl	1.719161	.7072902	1.32	0.188	.7675715	3.850476
ht	6.249602	4.102026	2.79	0.005	1.726445	22.6231
ui	2.1351	1.042775	1.55	0.120	.8197749	5.560858
_cons	1.586014	1.939482	0.38	0.706	.144345	17.42658

Note: **_cons** estimates baseline odds.

Also, you can specify `vce(cluster` *clustvar*`)` and then, within cluster, relax the assumption of independence. To illustrate this, we have made some fictional additions to the low-birthweight data.

Say that these data are not a random sample of mothers but instead are a random sample of mothers from a random sample of hospitals. In fact, that may be true—we do not know the history of these data.

Hospitals specialize, and it would not be too incorrect to say that some hospitals specialize in more difficult cases. We are going to show two extremes. In one, all hospitals are alike, but we are going to estimate under the possibility that they might differ. In the other, hospitals are strikingly different. In both cases, we assume that patients are drawn from 20 hospitals.

In both examples, we will fit the same model, and we will type the same command to fit it. Below are the same data we have been using but with a new variable, `hospid`, that identifies from which of the 20 hospitals each patient was drawn (and which we have made up):

```
. use https://www.stata-press.com/data/r19/hospid1, clear
. logistic low age lwt i.race smoke ptl ht ui, vce(cluster hospid)
Logistic regression                                     Number of obs =    189
                                                        Wald chi2(8)  =  49.67
                                                        Prob > chi2   = 0.0000
Log pseudolikelihood = -100.724                         Pseudo R2     = 0.1416
                                (Std. err. adjusted for 20 clusters in hospid)
------------------------------------------------------------------------------
             |               Robust
         low | Odds ratio   std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         age |   .9732636   .0397476    -0.66   0.507      .898396    1.05437
         lwt |   .9849634   .0057101    -2.61   0.009     .9738352    .9962187
             |
        race |
       Black |   3.534767   2.013285     2.22   0.027     1.157563    10.79386
       Other |   2.368079   .8451325     2.42   0.016     1.176562    4.766257
             |
       smoke |   2.517698   .8284259     2.81   0.005     1.321062     4.79826
         ptl |   1.719161   .6676221     1.40   0.163     .8030814    3.680219
          ht |   6.249602   4.066275     2.82   0.005      1.74591    22.37086
          ui |     2.1351   1.093144     1.48   0.138     .7827337    5.824014
       _cons |   1.586014   1.661913     0.44   0.660     .2034094    12.36639
------------------------------------------------------------------------------
Note: _cons estimates baseline odds.
```

The standard errors are similar to the standard errors we have previously obtained, whether we used the robust or conventional estimators. In this example, we invented the hospital IDs randomly.

Here are the results of the estimation with the same data but with a different set of hospital IDs:

```
. use https://www.stata-press.com/data/r19/hospid2
. logistic low age lwt i.race smoke ptl ht ui, vce(cluster hospid)
Logistic regression                                     Number of obs =    189
                                                        Wald chi2(8)  =   7.19
                                                        Prob > chi2   = 0.5167
Log pseudolikelihood = -100.724                         Pseudo R2     = 0.1416
                                (Std. err. adjusted for 20 clusters in hospid)
------------------------------------------------------------------------------
             |               Robust
         low | Odds ratio   std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         age |   .9732636   .0293064    -0.90   0.368     .9174862    1.032432
         lwt |   .9849634   .0106123    -1.41   0.160     .9643817    1.005984
             |
        race |
       Black |   3.534767   3.120338     1.43   0.153     .6265521     19.9418
       Other |   2.368079   1.297738     1.57   0.116     .8089594    6.932114
             |
       smoke |   2.517698   1.570287     1.48   0.139     .7414969    8.548655
         ptl |   1.719161   .6799153     1.37   0.171     .7919045    3.732161
          ht |   6.249602   7.165454     1.60   0.110      .660558    59.12808
          ui |     2.1351   1.411977     1.15   0.251     .5841231    7.804266
       _cons |   1.586014   1.946253     0.38   0.707     .1431423      17.573
------------------------------------------------------------------------------
Note: _cons estimates baseline odds.
```

Note the strikingly larger standard errors. What happened? In these data, women most likely to have low-birthweight babies are sent to certain hospitals, and the decision on likeliness is based not just on age, smoking history, etc., but on other things that doctors can see but that are not recorded in our data. Thus, merely because a woman is at one of the centers identifies her to be more likely to have a low-birthweight baby.

Video examples

Logistic regression, part 1: Binary predictors

Logistic regression, part 2: Continuous predictors

Logistic regression, part 3: Factor variables

Stored results

`logistic` stores the following in `e()`:

Scalars

`e(N)`	number of observations
`e(N_cds)`	number of completely determined successes
`e(N_cdf)`	number of completely determined failures
`e(k)`	number of parameters
`e(k_eq)`	number of equations in `e(b)`
`e(k_eq_model)`	number of equations in overall model test
`e(k_dv)`	number of dependent variables
`e(df_m)`	model degrees of freedom
`e(r2_p)`	pseudo-R^2
`e(ll)`	log likelihood
`e(ll_0)`	log likelihood, constant-only model
`e(N_clust)`	number of clusters
`e(chi2)`	χ^2
`e(p)`	p-value for model test
`e(rank)`	rank of `e(V)`
`e(ic)`	number of iterations
`e(rc)`	return code
`e(converged)`	`1` if converged, `0` otherwise

Macros

`e(cmd)`	`logistic`
`e(cmdline)`	command as typed
`e(depvar)`	name of dependent variable
`e(wtype)`	weight type
`e(wexp)`	weight expression
`e(title)`	title in estimation output
`e(clustvar)`	name of cluster variable
`e(offset)`	linear offset variable
`e(chi2type)`	`Wald` or `LR`; type of model χ^2 test
`e(vce)`	*vcetype* specified in `vce()`
`e(vcetype)`	title used to label Std. err.
`e(opt)`	type of optimization
`e(which)`	`max` or `min`; whether optimizer is to perform maximization or minimization
`e(ml_method)`	type of `ml` method
`e(user)`	name of likelihood-evaluator program
`e(technique)`	maximization technique
`e(properties)`	`b V`
`e(estat_cmd)`	program used to implement `estat`
`e(predict)`	program used to implement `predict`
`e(marginsok)`	predictions allowed by `margins`
`e(marginsnotok)`	predictions disallowed by `margins`
`e(asbalanced)`	factor variables `fvset` as `asbalanced`
`e(asobserved)`	factor variables `fvset` as `asobserved`

Matrices

`e(b)`	coefficient vector
`e(Cns)`	constraints matrix
`e(ilog)`	iteration log (up to 20 iterations)
`e(gradient)`	gradient vector
`e(mns)`	vector of means of the independent variables
`e(rules)`	information about perfect predictors
`e(V)`	variance–covariance matrix of the estimators
`e(V_modelbased)`	model-based variance

Functions

`e(sample)`	marks estimation sample

In addition to the above, the following is stored in r():

Matrices

r(table)	matrix containing the coefficients with their standard errors, test statistics, p-values, and confidence intervals

Note that results stored in r() are updated when the command is replayed and will be replaced when any r-class command is run after the estimation command.

Methods and formulas

Define $\mathbf{x}_j$ as the (row) vector of independent variables, augmented by 1, and $\mathbf{b}$ as the corresponding estimated parameter (column) vector. The logistic regression model is fit by logit; see [R] **logit** for details of estimation.

The odds ratio corresponding to the ith coefficient is $\psi_i = \exp(b_i)$. The standard error of the odds ratio is $s_i^{\psi} = \psi_i s_i$, where s_i is the standard error of b_i estimated by logit.

Define $I_j = \mathbf{x}_j\mathbf{b}$ as the predicted index of the jth observation. The predicted probability of a positive outcome is

$$p_j = \frac{\exp(I_j)}{1 + \exp(I_j)}$$

This command supports the Huber/White/sandwich estimator of the variance and its clustered version using vce(robust) and vce(cluster *clustvar*), respectively. See [P] **_robust**, particularly *Maximum likelihood estimators* and *Methods and formulas*.

logistic also supports estimation with survey data. For details on VCEs with survey data, see [SVY] **Variance estimation**.

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Also see

[R] **logistic postestimation** — Postestimation tools for logistic

[R] **brier** — Brier score decomposition

[R] **cloglog** — Complementary log–log regression

[R] **exlogistic** — Exact logistic regression

[R] **logit** — Logistic regression, reporting coefficients

[R] **npregress kernel** — Nonparametric kernel regression

[R] **npregress series** — Nonparametric series regression

[R] **roc** — Receiver operating characteristic (ROC) analysis

[BAYES] **bayes: logistic** — Bayesian logistic regression, reporting odds ratios

[FMM] **fmm: logit** — Finite mixtures of logistic regression models

[LASSO] **Lasso intro** — Introduction to lasso

[MI] **Estimation** — Estimation commands for use with mi estimate

[PSS-2] **power logistic general** — Power analysis for logistic regression: General case+

[PSS-2] **power logistic onebin** — Power analysis for logistic regression with one binary covariate+

[PSS-2] **power logistic twobin** — Power analysis for logistic regression with two binary covariates+

[SVY] **svy estimation** — Estimation commands for survey data

[XT] **xtlogit** — Fixed-effects, random-effects, and population-averaged logit models

[U] 20 Estimation and postestimation commands

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